Please write clearly in block capitals. Centre number | Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE

Paper 2 (Generated practice paper - Set 07)

Assessed: The living environment, Biological resources, Sustainability, Research methods

Time allowed: 3 hours

Materials

For this paper you may use:
- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
  - use good English
  - organise information clearly
  - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
2	
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9	
10	
11	
TOTAL	

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Answer all questions in the spaces provided.

0 1 . 1 Explain the difference between primary and secondary succession and why secondary succession is faster.

[3 marks]

0 1 . 2 Explain what is meant by **deflected succession** and a **plagioclimax**, using a named UK example.

[3 marks]

0 1 . 3 Explain why conservation of a plagioclimax requires continued human activity whereas conservation of a rainforest or coral reef does not.

[2 marks]

0 1 . 4 Explain why biodiversity tends to be low in abiotically extreme environments and high in stable ones.

[2 marks]

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0 2 . 1 Give **two** habitats created by farming that support important wildlife communities.

[2 marks]

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0 2 . 2 Explain how drainage and nutrient enrichment of farmland reduce biodiversity.

[2 marks]

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0 2 . 3 Explain **one** problem that can be caused by introducing a biological control species.

[1 mark]

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0 3 . 1 Explain how each of the following new technologies improves the validity of ecological research.

[8 marks]

Satellite / radio / GPS tracking

DNA databases and eDNA

Image recognition software

Acoustic monitoring / sonograms

0 3 . 2 Explain why new technologies allow the collection of 'more representative' data.

[2 marks]

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0 4 The Isle of Eigg (area 31 km^2, population about 100) is not connected to the national grid. Table 1 shows its electricity sources.

Table 1

Source	Capacity / kW
Hydroelectric (3 generators)	110
Solar PV	50
Wind (4 turbines)	24
Diesel generators (back-up)	160

0 4 . 1 Explain why Eigg uses a mix of three renewable resources rather than one, and why diesel back-up is still needed.

[3 marks]

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0 4 . 2 Describe **three** other ways in which the Eigg community has increased its sustainability.

[3 marks]

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0 4 . 3 Explain why small self-contained communities often find it easier to adopt sustainable lifestyles than large cities.

[2 marks]

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0 4 . 4 Explain how large-scale DC electricity grids such as Desertec could increase the use of renewable energy.

[2 marks]

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0 5 . 1 Explain why **proxy data** are needed to study the early conditions on Earth.

[1 mark]

0 5 . 2 Complete Table 1 to show what each analytical technique can be used to estimate.

[4 marks]

Table 1

Technique	What it estimates	
Ratio of carbon-12 to carbon-14 in a sample		
Ratio of oxygen-18 to oxygen-16		
Analysis of air bubbles trapped in ice cores		
Ratio of magnesium to calcium in calcite deposits		

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0 5 . 3 Explain why an understanding of continental drift and ocean currents has been important in explaining past changes in conditions.

[2 marks]

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0 5 . 4 Life probably first developed around volcanic vents on the seabed. Explain why the conditions in the oceans were more suitable than those on land 3.5 billion years ago.

[3 marks]

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0 6 The Russell formula describes changes in a fish stock: $S_2 = S_1 + (A + G) - (C + M)$, where S_1 and S_2 are the stock biomass at the start and end of the year, A = recruitment of young fish, G = growth, C = catch and M = natural mortality. Table 1 gives estimates for a cod stock.

Table 1

Term	Biomass / thousand tonnes
S_1	600
A	90
G	150
C	220
M	80

0 6 . 1 Calculate S_2 and state whether the stock is being fished sustainably.

[2 marks]

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Answer [ ] thousand tonnes

0 6 . 2 Calculate the maximum catch that would have kept the stock constant.

[1 mark]

Answer [ ] thousand tonnes

0 6 . 3 Explain why the catch is the only term in the formula that is easy to measure, and why the others are difficult.

[3 marks]

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0 6 . 4 Describe how each of the following measures obtained from commercial catches can indicate overfishing.

[4 marks]

Catch per unit fishing effort .....

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Mean fish size and mean age .....

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0 7 . 1 Explain why the ecosystem services provided by other species are often given a low conservation priority.

[2 marks]

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0 7 . 2 Explain how soil organisms provide services that humans depend on.

[3 marks]

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0 7 . 3 Explain why pollination by insects is more useful than wind pollination to plants growing in forests.

[2 marks]

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0 7 . 4 Explain how seed dispersal by animals increases the survival of a plant's offspring.

[3 marks]

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0 8 Table 1 shows climate data for two sites where a farmer is considering growing maize, a frost-sensitive crop that needs a thermal growing season of at least 120 days.

Table 1

Feature	Site A: valley floor	Site B: south-facing slope
Frost-free period / days	110	150
Days with mean temperature above 5 °C	160	175
Mean summer rainfall / mm	180	150
Gradient / %	1	8

0 8 . 1 Use Table 1 to explain which site is more suitable for maize.

[3 marks]

0 8 . 2 Describe **three** ways in which temperature can be controlled to increase crop productivity.

[3 marks]

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0 8 . 3 Explain why keeping livestock warm increases the amount of food converted to growth.

[2 marks]

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0 8 . 4 Explain how day length affects livestock production and how it can be manipulated.

[2 marks]

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0 9 Table 1 shows typical energy values for an agroecosystem. Values are units of energy remaining from 100 000 units of solar energy.

Table 1

Stage	Energy / arbitrary units
Solar energy reaching the crop	100 000
Energy captured during photosynthesis	4 000
Energy stored in crop biomass	1 000

0 9 . 1 Calculate the percentage of the solar energy reaching the crop that is stored in crop biomass.

[1 mark]

Answer ☐ %

0 9 . 2 Explain why so little of the solar energy is stored in the crop.

[3 marks]

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0 9 . 3 Explain why few food chains have more than four trophic levels.

[2 marks]

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0 9 . 4 Explain why rearing sheep or cattle is the best option for food production in upland and semi-arid areas.

[2 marks]

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0 9 . 5 Explain how pigs can increase the efficiency of a food production system.

[2 marks]

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1 0 . 1 Describe the conservation efforts used to protect tropical rainforests and evaluate their effectiveness.

[9 marks]

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Extra space

1 0 . 2 State **one** reason why rainforest conservation is important to countries far from the tropics.

[1 mark]

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Write an essay on one of the following topics.

1 1 . 1 Using tropical coral reefs as an example, discuss how the ecological features of a habitat determine both its importance and its vulnerability to human activities.

[25 marks]

OR

1 1 . 2 Discuss how feedback mechanisms, tipping points and diversity influence the stability of natural systems, and how human activities affect them.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

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END OF QUESTIONS